

Asim Önder (歐雅森), Ph.D.

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Research Profile

I am an engineer and computational scientist with a focus on multiphase and turbulent flows. My research combines high-accuracy numerical algorithms, theoretical analysis, and data-driven modeling to investigate fundamental and practical problems in air-sea exchange, wave dynamics, and flow-structure interaction.

Education

Ph.D., Mechanical Engineering <i>KU Leuven, Leuven, Belgium</i>	Dec 2014
M.Sc., Computational Science and Engineering <i>Technical University of Munich, Munich, Germany</i>	May 2009
B.Sc., Mechanical Engineering <i>Istanbul Technical University, Istanbul, Turkey</i>	Jun 2006

Professional Experience

Assistant Professor <i>National Sun Yat-sen University, Kaohsiung, Taiwan</i> <i>Department of Marine Environment and Engineering</i>	Aug 2022 – Present
Senior Research Fellow <i>National University of Singapore, Singapore</i> <i>Department of Civil and Environmental Engineering</i>	Jan 2022 – July 2022
Research Fellow <i>National University of Singapore, Singapore</i> <i>Department of Civil and Environmental Engineering</i>	Oct 2015 – Dec 2021
Postdoctoral Researcher <i>KU Leuven, Leuven, Belgium</i> <i>Department of Mechanical Engineering</i>	Jan 2015 – Oct 2015

Affiliated Appointments

Affiliated Researcher <i>Texas A&M University, College Station, TX, USA</i> <i>Department of Civil & Environmental Engineering</i>	May 2025 – May 2026
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Faculty Sponsors: Prof. Philip L.-F. Liu & Prof. Kuang-An Chang.

Research Grants as Principal Investigator

- **Exchange Processes in Marine Surface Layers: A Fully-Coupled Approach (II)**

Agency: National Science and Technology Council (NSTC), Taiwan

Period: Aug 2024 – Jul 2027

Amount: 3,322,000 NT\$ (approx. €96,000 EUR)

- **Air-Sea Interactions in Marine Surface Layers: A Fully-Coupled Approach**

Agency: NSTC, Taiwan

Period: Mar 2022 – Jul 2024

Amount: 800,000 NT\$ (approx. €23,000 EUR)

- **Simulation of Air-Sea Interactions with AI-Accelerated CFD**

Agencies: NSCC Singapore & Riken Japan

Grant: 315,654 node hours on the Fugaku supercomputer.

Period: Apr 2022 – Mar 2023

Peer-Reviewed Publications

1. Tan, W., Yuan, J., Cao, D., & **Önder, A.** (2025). Experiments and large eddy simulations of oscillatory flow over vortex ripples at high Reynolds number. *Coastal Engineering*, 104881. [DOI]
2. **Önder, A.**, & Liu, P. L.-F. (2023). Deep learning of interfacial normal and curvature: a symmetry-preserving approach for the volume of fluid method. *Journal of Computational Physics*, 485, 112110. [DOI]
3. Goit, J. P., & **Önder, A.** (2022). The effect of coastal terrain on nearshore offshore wind farms: A large-eddy simulation study. *Journal of Renewable and Sustainable Energy*, 14(4). [DOI]
4. **Önder, A.**, & Liu, P. L.-F. (2021). Receptivity and transition in a solitary wave boundary layer over rough bottom topography. *Journal of Fluid Mechanics*, 912, A21. [DOI]
5. **Önder, A.**, & Liu, P. L.-F. (2020). Stability of the solitary wave boundary layer subject to finite-amplitude disturbances. *Journal of Fluid Mechanics*. 896, A20. [DOI]
6. **Önder, A.**, & Yuan J. (2019). Turbulent dynamics of sinusoidal oscillatory flow over a wavy bottom. *Journal of Fluid Mechanics*. 858, 264-314. [DOI]
7. **Önder, A.**, & Meyers, J. (2018). On the interaction of very-large-scale motions in a neutral atmospheric boundary layer with a row of wind turbines. *Journal of Fluid Mechanics*. 841, 1040-1072. [DOI]
8. **Önder, A.**, & Meyers, J. (2016). Optimal control of a transitional jet using a continuous adjoint method. *Computers & Fluids*, 126, 12-24. [DOI]
9. **Önder, A.**, & Meyers, J. (2014). Modification of vortex dynamics and transport properties of transitional axisymmetric jets using zero-net-mass-flux actuation. *Physics of Fluids*, 26(7). [DOI]

Invited Talks

- “Air–Sea Interaction under Light Winds: Wind Stress and Short-Wave Dynamics.” *Invited Seminar, Institute of Oceanography, National Taiwan University*, Taipei, Taiwan, November 2025.
- “Wind-Sea Coupling: Perspectives on Momentum Transfer from High-Fidelity Simulation.” *Invited Seminar, Department of Engineering Science and Ocean Engineering, National Taiwan University*, Taipei, Taiwan, September 2025.

- “High-Fidelity Two-Phase Flow Simulation of Turbulent Air–Sea Exchange using Sharp Interface Methods.” *Keynote Speech, 30th National Computational Fluid Dynamics Conference*, Tainan, Taiwan, Aug 2025.
- “Neural-Network Models for Curvature Estimation in Two-Phase Interfacial Flows.” *Taiwan Society for Industrial and Applied Mathematics Annual Meeting*, Taichung, Taiwan, May 2024.
- “High-Fidelity Simulations of Oceanic Flows: Tsunamis, Turbulence and Machine Learning.” *Invited Seminar, Institute of Hydrological and Oceanic Sciences, National Central University*, Taoyuan, Taiwan, Dec 2023.
- “Resolving Tsunami Turbulence: Scientific and Computational Challenges.” *State Grid HPC/AI User Achievements Exchange Conference*, Tainan, Taiwan, Dec 2023.
- “On Bottom Drag and Turbulence Under Tsunami-Like Long Waves.” *Invited Seminar, Department of Hydraulic and Ocean Engineering, National Cheng Kung University*, Tainan, Taiwan, Jan 2023.
- “Turbulent Boundary Layers Beneath Tsunami-Scale Long Waves.” *1st Taiwan Society of Fluid Dynamics Conference*, Hsinchu, Taiwan, Dec 2022.
- “Can Tsunamis Generate Turbulence in Deep Waters?” *44th Ocean Engineering Conference*, Kaohsiung, Taiwan, Nov 2022.
- “Towards Fully Resolving the Turbulence Around Wave-Induced Bedforms Using Petascale Supercomputing.” *Supercomputing Frontiers 2017*, Singapore, Mar 2017.

Teaching Experience

Instructor of Record, National Sun Yat-sen University 2022 – Present

- MAEV204: Engineering Mathematics II (Undergraduate, Required, 3 offerings)
- MAEV522: Marine System Modelling I (Graduate, Elective, 3 offerings)
- MAEV525: Applied Engineering Hydraulics (Graduate, Elective, 3 offerings)
- MAEV628: Turbulence (Graduate, Elective, 2 offerings)
- MAEV240: Engineering Mechanics (Undergraduate, Required, 1 offering)
- MAEV504: Seminar in Marine Environment (Graduate, Required, 1 offering)

Awards and Honors

- **Excellent Teaching Course Award (Engineering Mathematics II)** Spring 2025
(Awarded for the 113-2 Academic Semester)
- **New Faculty Award**, National Sun Yat-sen University Aug 2024
(Awarded for the 113 Academic Year)
- **Excellent Teaching Course Award (Engineering Mathematics II)** Spring 2024
(Awarded for the 112-2 Academic Semester)
- **New Faculty Award**, National Sun Yat-sen University Aug 2022
(Awarded for the 111 Academic Year)
- **Excellent Teaching Course Award (Engineering Mechanics)** Spring 2023
(Awarded for the 111-2 Academic Semester)

Professional Service

Peer Reviewer for *Journal of Fluid Mechanics*, *Journal of Computational Physics*, *Journal of Geophysical Research*, *International Journal of Multiphase Flow*

Professional Memberships include the American Physical Society (APS) and the European Geosciences Union (EGU).